

Efficient Medical Image Compression ...

Team 8 By:

Haarini SK-22MID0231

Sivaranjini TS-22MID0232

Drishna B-22MID0273

Efficient Medical Image Compression Using Discrete Wavelet Transform

A Performance Analysis Approach



Specificity: Clearly identifies the compression technique (DWT) and application domain (medical imaging)



Scope Definition: Emphasizes performance analysis as the primary research methodology



Academic Relevance: Addresses healthcare technology challenges of storage and transmission efficiency



Technical Focus: Highlights the discrete wavelet transform as the core algorithmic approach

Problem Statement: Healthcare Imaging Challenges

Current Healthcare Imaging Challenges:



Medical imaging generates massive data volumes, creating storage and transmission bottlenecks.



High-resolution images range from 8-16 MB, with some exceeding 100 MB.



Thousands of images per day cause exponential storage growth.

Critical Issues Addressed:



Storage infrastructure strain



Diagnostic quality vs. efficiency trade-off



Network transmission delays



Regulatory compliance complexity

Technical Gap: JPEG/JPEG2000 fail to preserve diagnostic details while reducing size

Research Objective: DWT-Based Medical Image Compression

Primary and Specific Goals



Primary Goal: Develop an optimized DWT-based medical image compression framework

- 1 Adaptive DWT-based algorithm for medical images
- 2 is the benchmark against JPEG, JPEG2000, DICOM
- 3 Ensure quality via PSNR, SSIM, diagnostic accuracy
- 4 Achieve 10:1 to 20:1 compression ratios without losing diagnostic quality.

Expected Outcomes and Success Metrics



Expected Outcomes: Novel DWT optimization, reduced costs, better speed, clinical-ready solution



Success Metrics: Compression ratio, quality preservation, speed, compatibility





Context & Motivation: Medical Imaging Data

Challenges in Healthcare Systems

-  Medical imaging data growth challenges healthcare systems
-  High-resolution images require heavy storage and network resources
-  Traditional compression compromises diagnostic accuracy
-  Need for healthcare-specific compression where accuracy is critical

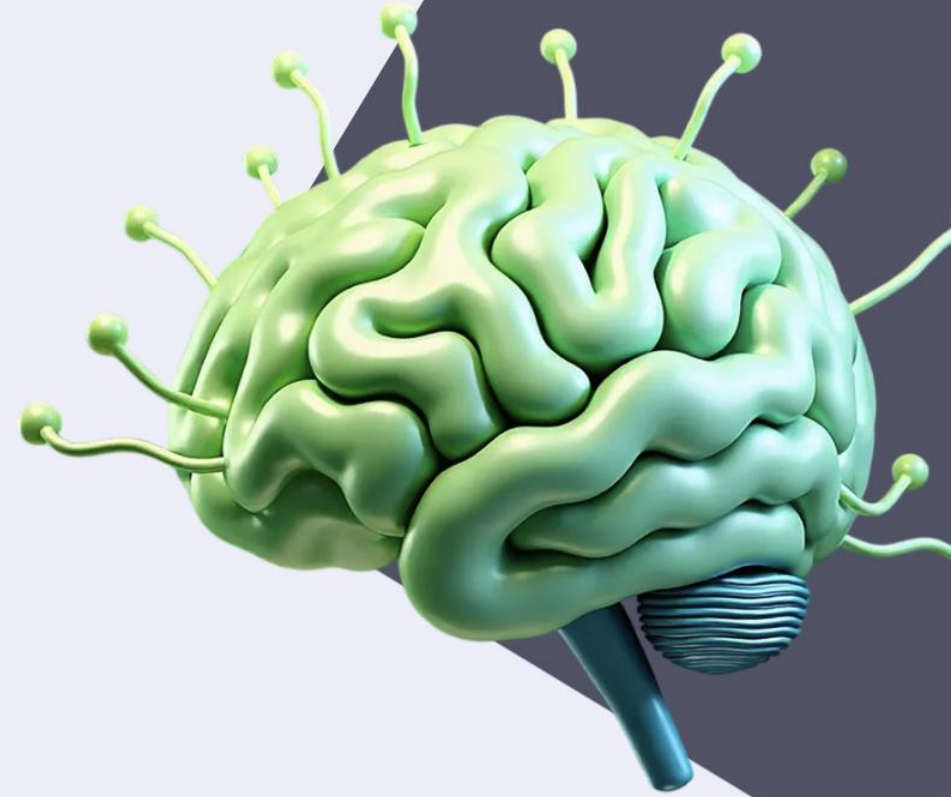
Abstract — Methodology & Innovation

Methodology:

-  Develop adaptive DWT framework
-  Multi-resolution analysis for feature preservation
-  Integrate medical quality metrics

Innovation:

-  Adaptive wavelet selection based on image content
-  Diagnostic region preservation algorithm
-  Multi-dimensional quality evaluation beyond PSNR/ SSIM



Abstract — Results & Impact

Performance

-  Compression ratios: 12: 1 to 25:1 with diagnostic quality
-  Superior edge and contrast preservation
-  Outperforms JPEG/JPEG2000/DICOM

Impact

-  Reduces healthcare IT costs
-  Enables faster telemedicine and better access in low-resource areas
-  Lays groundwork for future medical image processing standards





Thank You